

F24-066-D-LearnScape

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Chapter 1

Introduction

This document outlines the basic requirements and design elements of our VR application, which intends to provide an immersive and interactive learning experience to its users. The application offers multiple features such as Real time QnA, Interactive Quizzes, Virtual Notes taking and Immersive Classroom environment. The document is intended to be read by developers, Project Managers , Marketing Staff and Users.

1.1 Problem Statement

There is an abundance of educational videos that you can find online today. After COVID-19 the popularity of online education bloomed and many people found themselves learning from educational content present online, however there are a couple of problems present in those videos.

Firstly, that these videos are not immersive enough, users don't get the feeling that they are sitting in a real classroom and attending the lecture. Which might hinder there understanding and concentration. Furthermore, the videos that are currently present on various educational platforms are not interactive enough. They don't make the viewer feel like he is a part of the lecture, which in turn hinders the user experience and affects the learning of the viewer.

There is no educational platform out there right now in which the user can interact with the content in Realtime and ask questions from the teacher and receive relevant feedback in Realtime. Our project addresses all of these issues through immersive Virtual Reality (VR) technology and advanced AI techniques. We plan to bridge this gap that is currently present in online educational videos.

1.2 Scope

The scope of this project is that we are developing a Virtual Reality (VR) application that changes the way current educational videos are perceived , by creating an immersive and interactive educational environment.

The main functionalities of our project includes :

Creating an Immersive VR Classroom Environment - It will make users feel like they are sitting in a real life classroom.

Realistic Teacher Motions - The virtual teacher will make actions just like a real teacher while teaching.

Realistic Teacher Emotions - The virtual teacher will make emotions just like a real teacher while teaching.

Realtime Question and Answers - The user will be able to stop the virtual teacher in the middle of the lecture and ask questions and will receive relevant feedback from the virtual teacher.

Lifelike content Delivery - The virtual teacher that will be delivering the lecture would feature a realistic voice to make the lecture feel more natural.

Dynamic slide management - The slides will automatically change according to what the teacher is currently saying to maintain context.

Note Taking Capability - In order to enhance the feel of a realistic classroom and to facilitate users we also have a virtual notepad in which users can take notes.

Interactive Quizzes - In order to enhance lecture retention and user engagement , we will also feature AI generated quizzes during our lectures.

Feedback System - We will take user feedback after each lecture , to keep on improving our application in the future.

User Authentication - For keeping the data of our users secure.

Multiplayer - Multiple students can attend the same lecture together in a virtual environment and can interact with each other

Live Teaching Sessions - A real life teacher can come in the place of our virtual teacher and teach the students in person.

For the current scope of this project we are only covering videos that are related to computer science and specifically those videos that are related to python (Beginner to Intermediate level), Furthermore currently we are only catering those videos in which the teacher is teaching using slides or is doing live coding. In the future we can also cater various different types of subjects and videos into our application aswell.

The project will also be implemented in a defined timeline and deliverables to ensure the effective development of this application.

1.3 Modules

Below mentioned are all of the modules and there respective features.

1.3.1 Immersive Classroom Environment

In this Module we are creating am Immersive 3D Classroom Environment for students, where they can interact with a virtual teacher and can engage with the study material like notes etc. Additionally, In this module we will also focus on adding animations and emotions to the virtual teacher to make the lecture feel more real.

1. Teacher Avatar Creation
2. Avatar Animation and Emotions
3. Virtual notes taking
4. 3D Classroom environment creation

1.3.2 Application Backend

In this module we will be mainly working on our applications backend , which involves fine tuning an AI model aswell as working on Speech to Text and Text to Speech conversion for Realtime QnA Functionality.

1. Text to Speech
2. Speech to Text
3. Data Collection and Processing
4. LLM Fine Tuning
5. LLM Model Deployment
6. Real time QnA
7. Quiz Generation

1.3.3 Multiplayer Functionality

In this module we going to add multiplayer functionality into our project. It will enhance our projects domain and will play a vital role of fostering interaction and engagement between multiple students.

1. Live teaching session
2. Multiple Students learning together (Group Study)

1.3.4 User Authentication

In this module we will be working on user authentication using Firebase authentication. This will ensure that only relevant people get access to the application and also keeps user data secure.

1. User Login / authentication

1.3.5 Feedback System

In this module we will be working on the feedback system , In which the users will be able to give feedback and the administrator will be able to view the feedback. This will allow us to keep on improving our application in the future.

1. User Feedback

1.4 User Classes and Characteristics

User class	Description
Student	The students or learners are the primary users of this application. They can attend lectures in a virtual classroom environment whilst interacting with many things such as Quizzes , virtual notes , virtual teacher model etc. They can Learn alone or with other students. They also have the ability to attend live teaching sessions where a real teacher comes in place of the virtual teacher model and conducts a lesson.
Teacher	The teacher is the one who is the main content creator , it is the content of the teacher that we will feature on our application. The teacher will provide us with the traditional video they want to transform along with the transcript and slides. The teacher can also engage with students in a live session , in which they can answer the students' questions themselves or teach any topic they like.
Administrator	The administrator can oversee the entire system , they can see all of the lectures that have been published on the application , along with all the students enrolled in the application. The administrator can also view any feedback which may have been given by any student regarding the application.

Table: User Classes and Characteristics

Chapter 2

Project Requirements

2.1 Use-cases

ID : UC01

Name : User Authentication

Primary Actor : User - Student , Teacher , Administrator

Description : The user will gain access to the application after logging in with correct credentials , each user will have their role specific functionalities.

ID : UC02

Name : Select Lecture

Primary Actor : Student

Description : The student will select which lecture he wants to watch from a list of available lectures , which will be separated into different modules. Each module will contain a specific number of lectures.

ID : UC03

Name : Join Virtual Classroom

Primary Actor : Student

Description : After the student selects a specific lecture , he will be placed inside a virtual classroom environment in which the lecture will take place.

ID : UC04

Name : Attend Lecture

Primary Actor : Student

Description : The student can take the virtual lecture , in which they can take notes , attempt Quizzes etc.

ID : UC05

Name : Take virtual Notes

Primary Actor : Student

Description : During the lecture , the student can take virtual notes , which can also be saved in the system to view later.

ID : UC06

Name : Attempt Virtual Quiz

Primary Actor : Student

Description : During the lecture , the student can attempt a virtual Quiz , which can improve their understanding of the topic.

ID : UC07

Name : Dynamic Slide Management

Primary Actor : System , Teacher

Description : The system will change the slides during the lecture according to the flow of the lecture.

ID : UC08

Name : Realistic Teacher Animations

Primary Actor : System

Description : The virtual model of the teacher will display realistic actions and emotions during the lecture , to improve the experience.

ID : UC09

Name : Ask Questions

Primary Actor : Student , System

Description : The student can ask questions from the virtual teacher during the lecture in real time.

ID : UC10

Name : Answer Questions

Primary Actor : System

Description : The system will generate an answer based on the students question , using our fine tuned AI model and will deliver the answer to the student in audio form through the virtual teachers model.

ID : UC11

Name : Feedback

Primary Actor : Student

Description : After each lecture the student has the option to give feedback regarding his experience of the application.

ID : UC12

Name : Multiplayer

Primary Actor : Student , Teacher

Description : Multiple students can attend a lecture together and interact with each other , creating a group learning environment. There is also the ability of the teacher conducting a live session to teach the students in person , instead of the virtual teacher model.

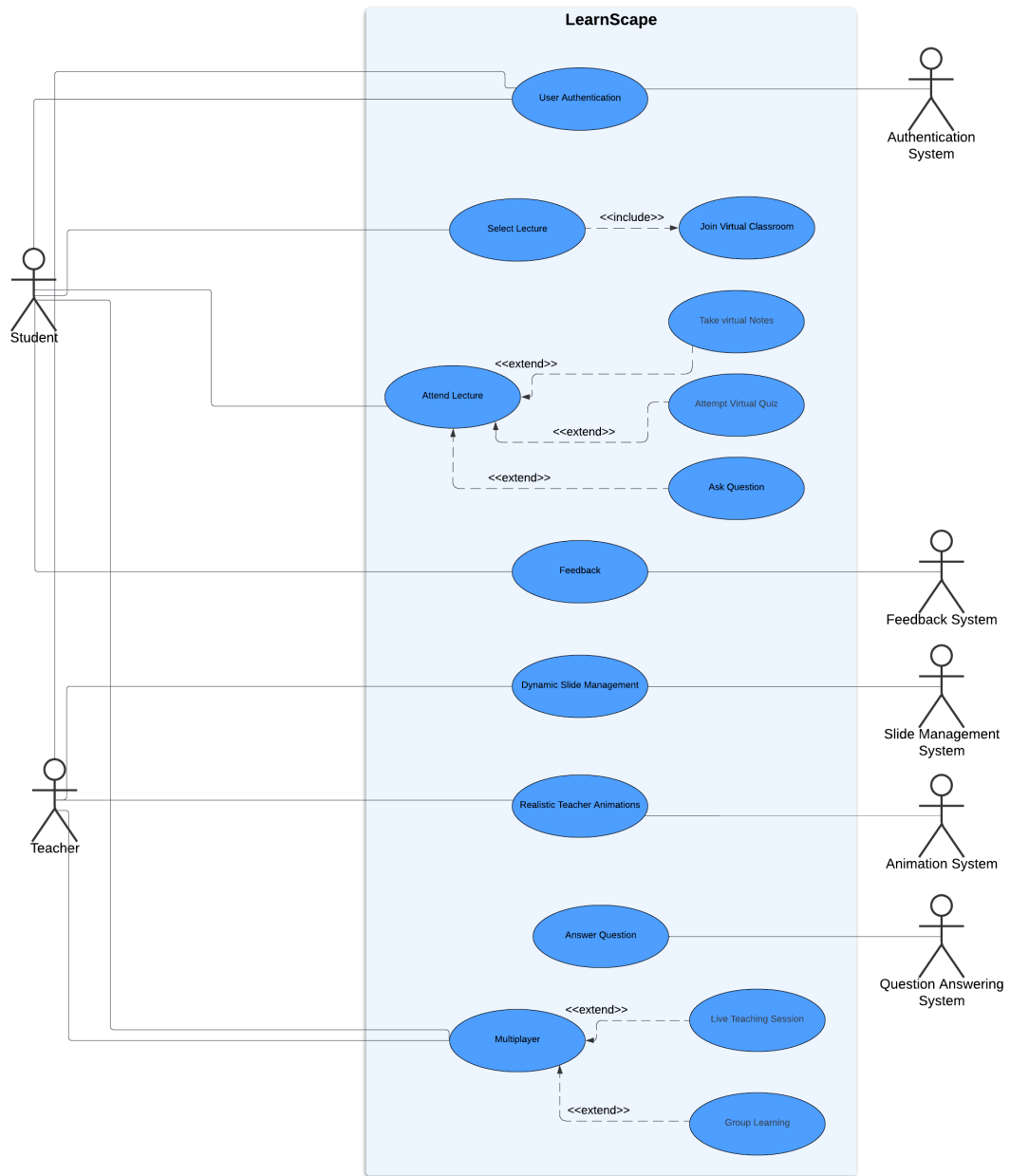


Figure 2.1: UseCase Diagram

2.2 Functional Requirements

2.2.1 User Authentication

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall allow users to create accounts by setting their own email and password.
2. **FR2** : The system will provide login authentication using Firebase.
3. **FR3** : The system will ensure that only authorized people gain access to the application.

2.2.2 Virtual Classroom Environment

Following are the functional requirements for the above mentioned feature:

1. **FR1** : System will allow users to join a virtual classroom environment in which they can view lectures.
2. **FR2** : System will provide an environment in which users can view the virtual teacher aswell as lecture content i.e slides.
3. **FR3** : System will work on synchronizing the slides with the teachers voice and current content being taught.
4. **FR4** : System Shall allow students to ask questions in Real Time during the lecture.
5. **FR5** : The system shall generate a relevant response for a student's question and deliver it to the student in the form of voice.

2.2.3 AI Generated Interactive Quizzes

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall allow users to attempt MCQ based AI generated Quizzes.
2. **FR2** : The system shall display quizzes to students after each lecture completion.
3. **FR3** : The system shall provide the students with instant results regarding performance.

2.2.4 Virtual Teacher Model

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall use a virtual teacher model to deliver the lecture.

2. **FR2** : The system shall feature a realistic voice for the teacher , during lecture delivery.
3. **FR3** : The system shall feature realistic actions and emotions for the teacher, during lecture delivery.
4. **FR4** : The system shall allow the virtual teacher model to answer students questions in real time using a fine tuned AI model.

2.2.5 Multiplayer

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall allow multiple students to attend lectures simultaneously.
2. **FR2** : The system shall allow multiple students to talk to each other for communication.
3. **FR3** : The system shall allow students to attend a live teaching session in which the teacher can teach the students in person.

2.2.6 Virtual Note Taking

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall allow students to take notes during the lecture.
2. **FR2** : The system shall allow the students to save their notes to view after the lecture.

2.2.7 User Feedback System

Following are the functional requirements for the above mentioned feature:

1. **FR1** : The system shall allow users to give feedback at the end of each lecture.
2. **FR2** : The system shall allow the administrator to view all the received feedback.

2.3 Non-Functional Requirements

Below mentioned are all of the non-functional requirements for our project :

2.3.1 Reliability

REL-1: The system shall have a mean time of failure of at least 500 hours

REL-2: In case of connection loss to the user , the user can join the session within 10 minutes.

REL-3: The system shall log all significant errors to the development team within 1 hour of detection , using a logging system.

2.3.2 Usability

USE-1: The user shall be able to join the classroom session in no more than 5 clicks , verified by UI.

USE-2: The system shall allow users to interact within the classroom with a VR controller within 10 minutes of use.

2.3.3 Performance

PER-1: The system shall load the virtual classroom environment before 5 minutes for users with internet connection with 20 mbps or more.

PER-2: The system shall handle 20 students simultaneously in a classroom during multiplayer.

PER-3: The system shall render classroom environment at a minimum 20 Fps on standard VR equipment.

2.3.4 Security

SEC-1: The system shall ensure that only authenticated users can gain access to the system using firebase authentication.

2.4 Domain Model

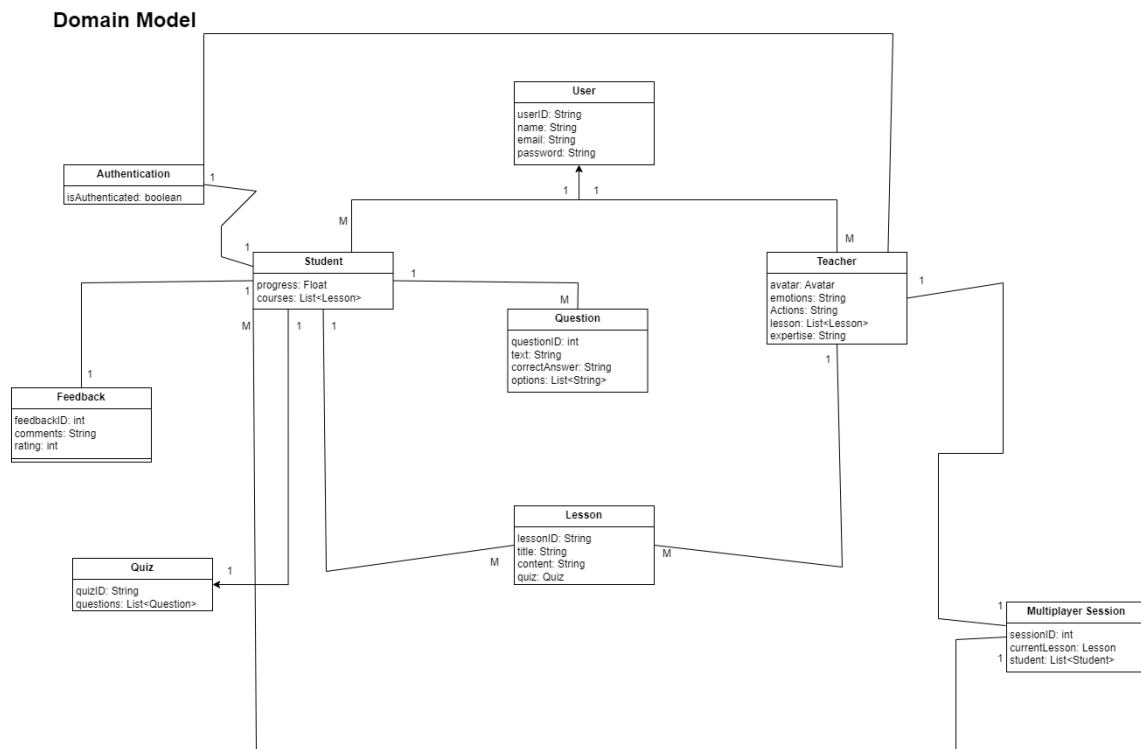


Figure 2.2: Domain Model Diagram

Chapter 3

System Overview

This section of the documentation will provide an overview of the functionality, context and design of our application. This part includes various diagrams such as activity diagram, class diagram, ERD etc., which show the our systems architecture and data relationships.

3.1 Architectural Design

The Architecture that our project LEARNSCAPE mainly follows is a client-server model. In which the unity client will request services from our central server. The user will interact with the interface and the different visuals through the Scene Editor Module and the Interaction Module which are present in the unity client.

The server will include multiple key components :

- Python Backend for processing
- LLM model for AI driven tasks
- PhotonPun for multiplayer functionality

The server side will also communicate with Firebase , which will act as the main database for our application, and also relies on Firebase authentication for secure user access.

The Front-end environment will basically be handles by the unity client which includes the Scene Editor and the Interaction modules. Meanwhile, all of the major processing including authentication , AI driven features , multiplayer will take place on the server side.

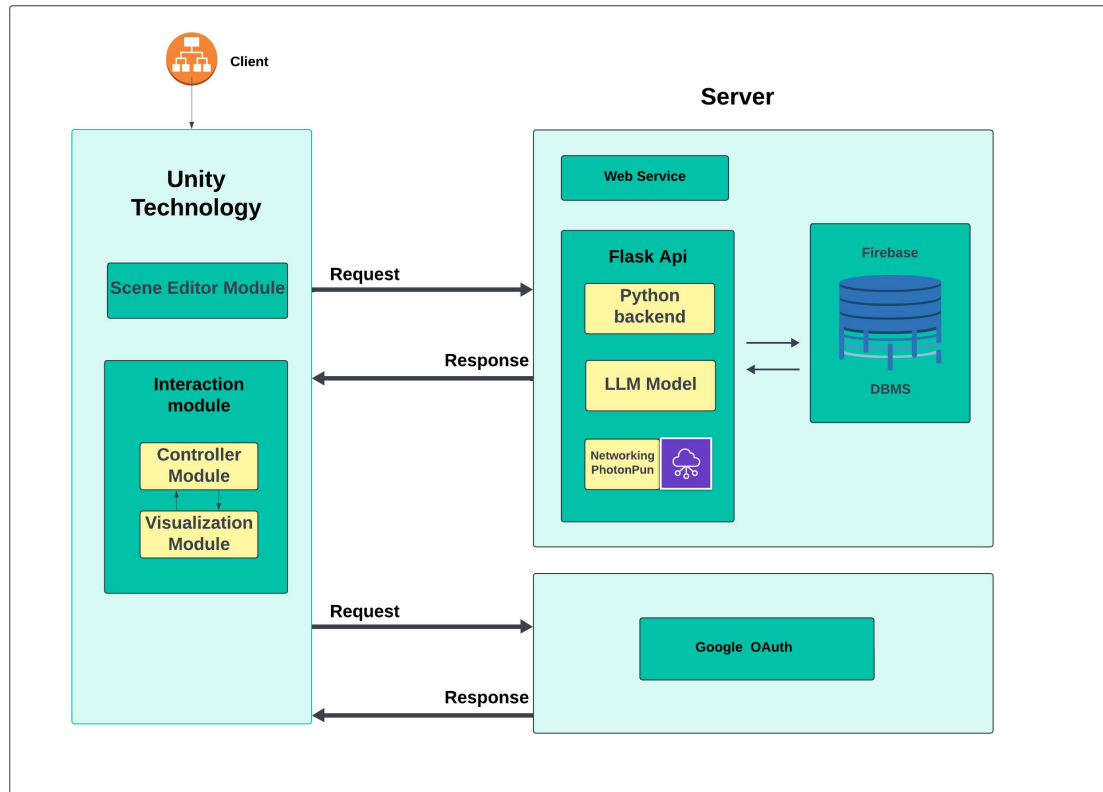


Figure 3.1: Architecture Diagram

3.2 Design Models

For our project we chose the Object Oriented development approach. Please find below all of the design models for the Object Oriented development approach :

3.2.1 Activity Diagram

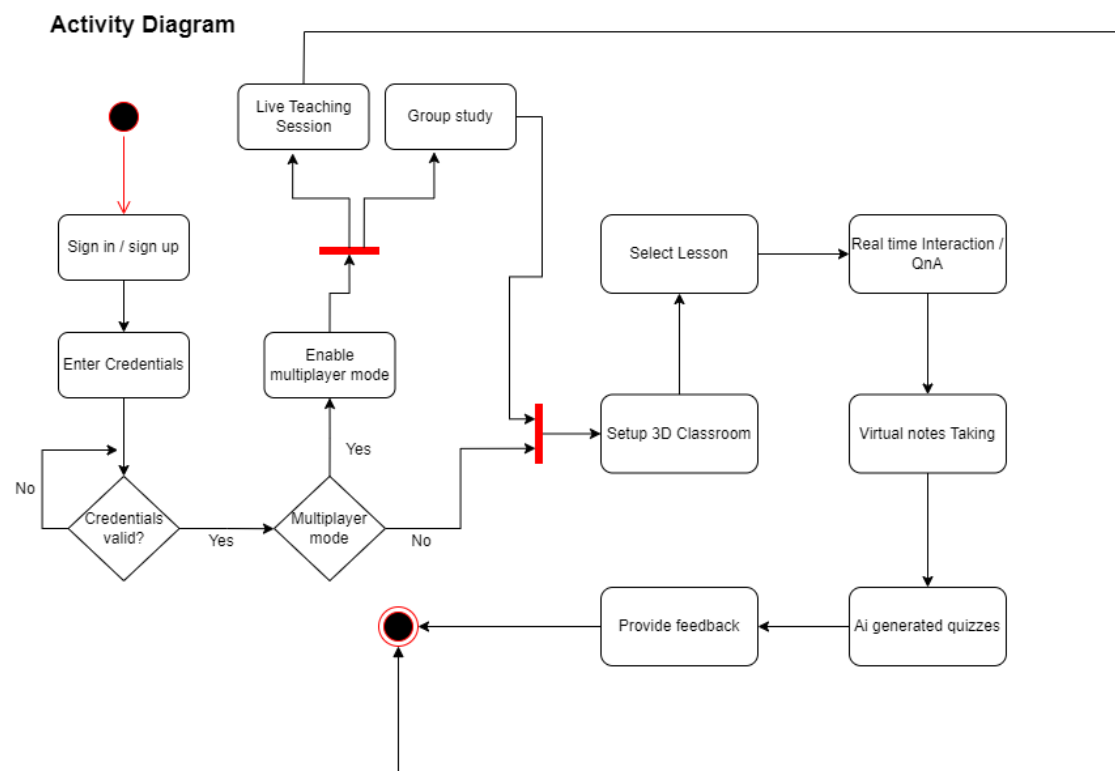


Figure 3.2: Activity Diagram

3.2.2 Class Diagram

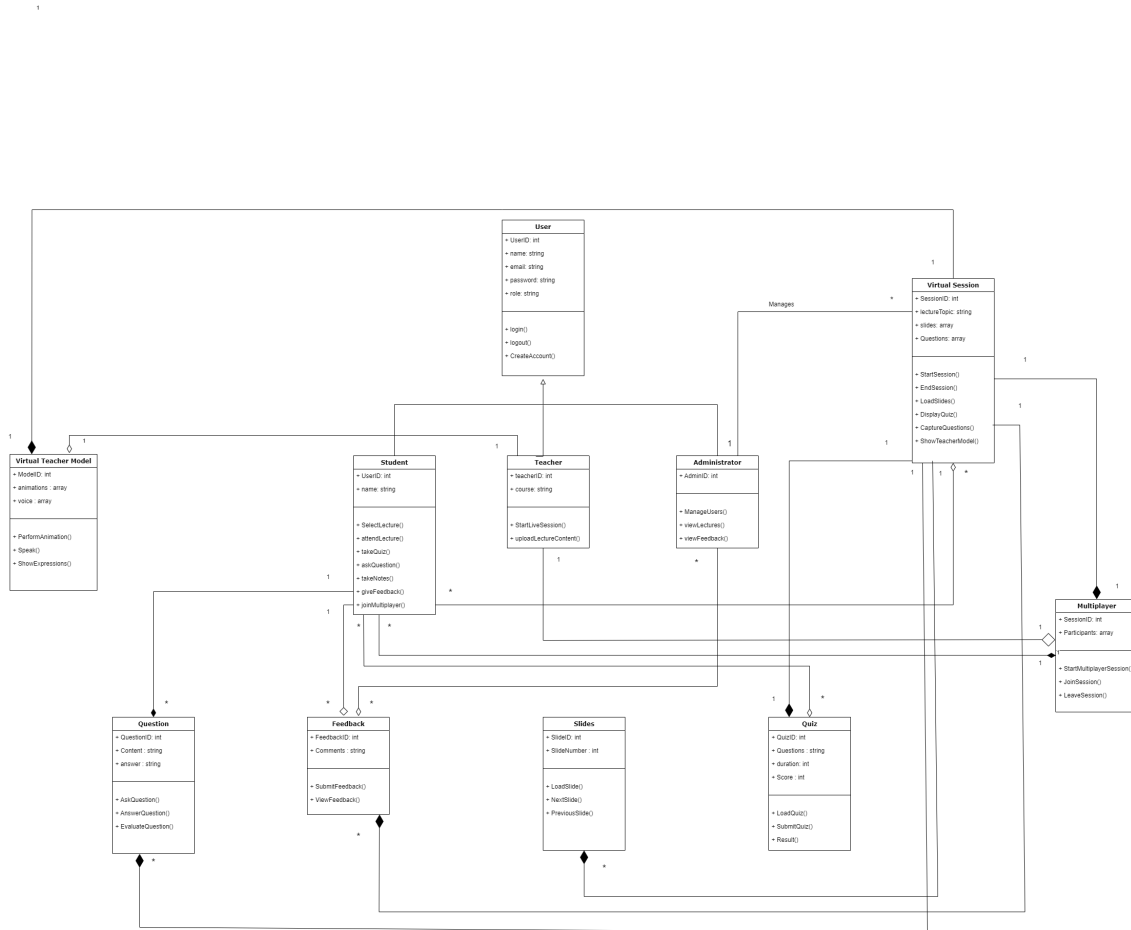


Figure 3.3: Class Diagram

3.2.3 Class Level Sequence Diagram

Class-level Sequence Diagram

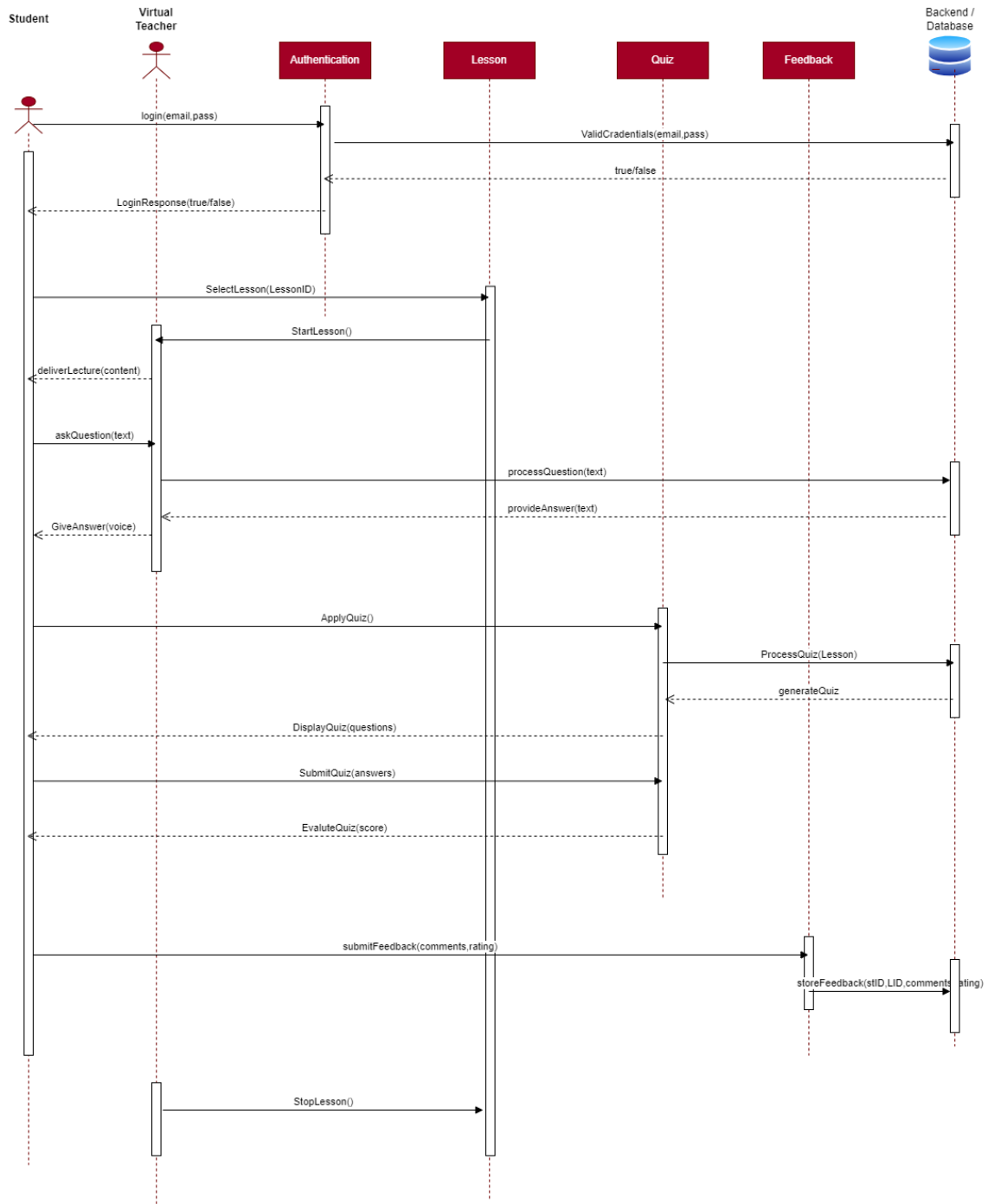


Figure 3.4: Sequence Diagram

3.2.4 State Transition Diagram

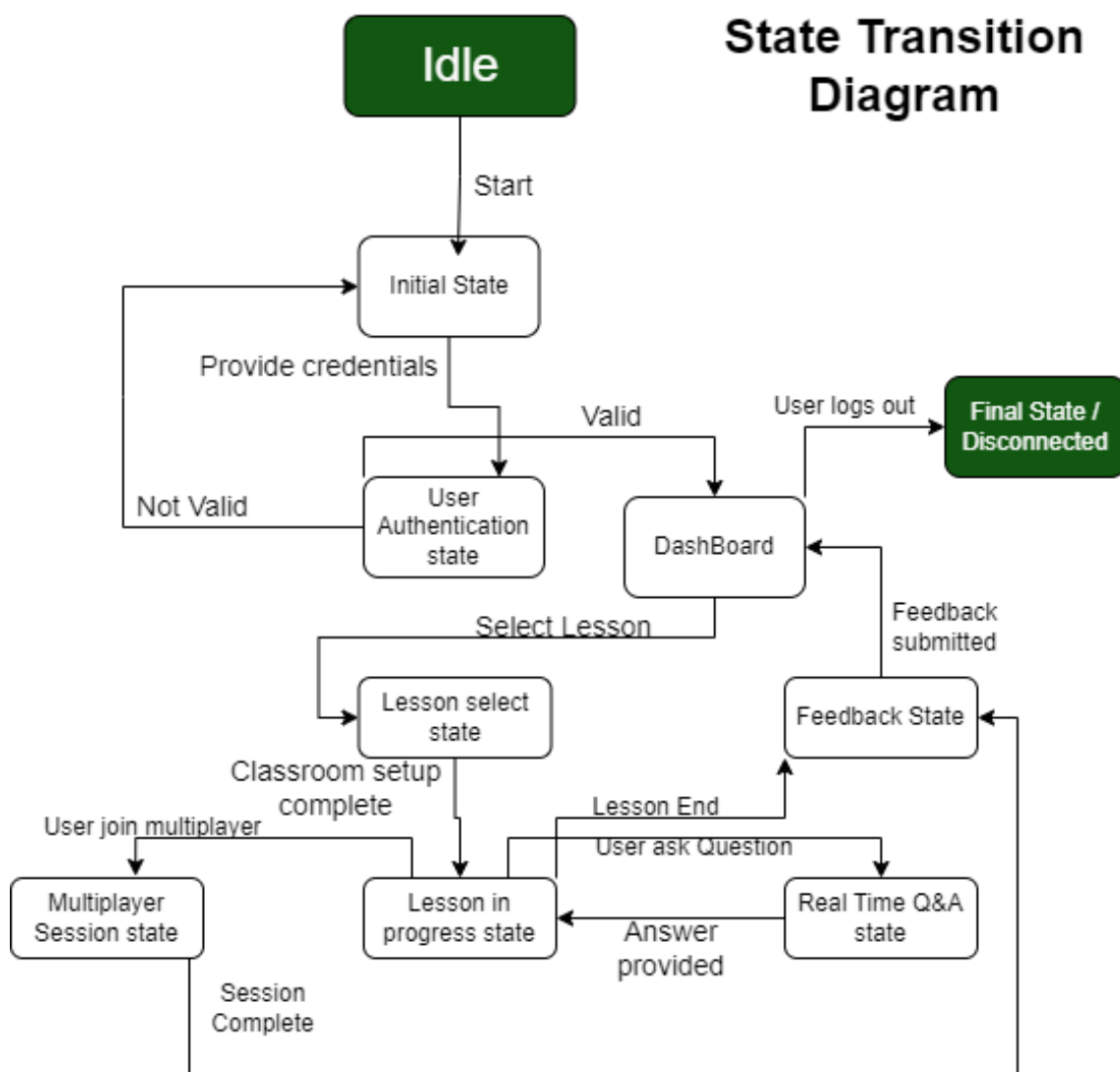


Figure 3.5: State Transition Diagram

3.3 Data Design

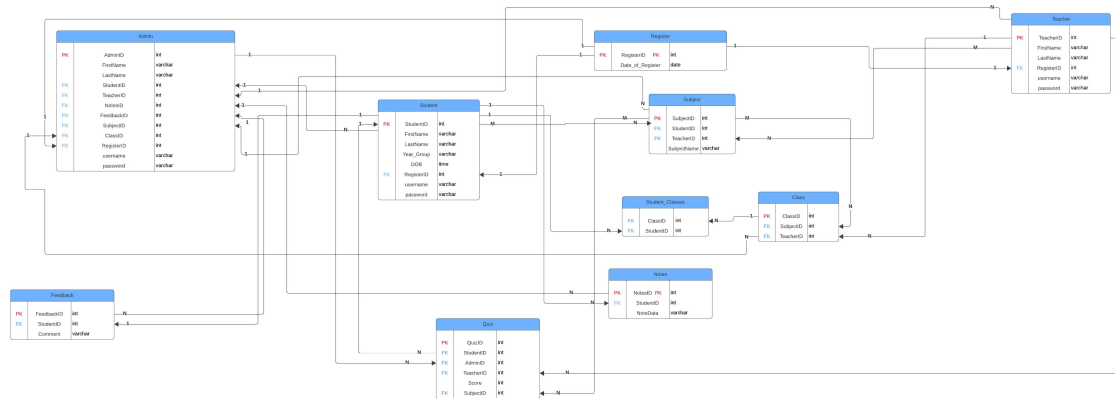


Figure 3.6: Entity Relationship Diagram

The main database which we will be using for our project would be Firebase. The above ERD represents the main data design of our application where Admin , Teacher , Subject , feedback , Notes and Quiz are the key entities.

- The data of all the Students , Teachers are stored as collections with some kind of unique identifiers in order to ensure uniqueness. All of these entities also have relationships with other entities like Quiz , subject etc.
- The Subject entity is also stored and linked through IDs to maintain relationships. Which ensures that we know which teacher is teaching which subject.
- entities like feedback , notes and quiz are also stored as separate collections and are linked backed to each student or teacher.
- The register entity also holds the registration ID of each user.

Bibliography